

Transfusion reaction algorithm

If there are clinical signs and symptoms of a transfusion reaction:

1. **STOP** transfusion immediately: do not disconnect.
2. **Assess** patient to determine if mild or moderate/severe reaction (see below), and check observations:
 - a. Call a MET if necessary: dial 55, code blue
 - b. Vital signs every 15 minutes until stable.
3. **Check** the patient ID, labels and forms for discrepancies.
4. **Inspect** unit for abnormal appearance (turbidity, clots).
5. **Do not** discard the blood product/pack and giving set.
6. **Notify** medical staff.
7. **Notify** transfusion lab: **SCGH** (638) 34018
OPH (645) 78088



Mild signs and symptoms

Febrile reaction?

- Isolated fever of $\geq 38^{\circ}\text{C}$ AND $\geq 1^{\circ}\text{C}$ above baseline
- Chills
- **No other signs or symptoms**

Allergic reaction?

- Localised rash
- **No other signs or symptoms**

- Consider symptomatic treatment.
 - If fever – antipyretics eg: 1g paracetamol
 - If rash – antihistamine eg: 10mg loratidine
- No blood tests required.
- If resuming transfusion, proceed cautiously. Direct nursing observation of patient for first 15 minutes and increased monitoring thereafter.

If the patient develops any other signs and symptoms
STOP THE TRANSFUSION
and follow the moderate/severe pathway.

Report

- Document in clinical notes.
- Send completed transfusion reaction form to transfusion laboratory SCGH, OPH.

Moderate to severe signs and symptoms

If one or more of the following signs/symptoms:

- Hypotension shock or hypertension
- Rigors or chills
- Pain: chest/back/abdominal/infusion site
- New/increasing shortness of breath/dyspnoea
- New wheeze
- New nausea/vomiting
- Haemoglobinuria (not haematuria)
- Temperature $\geq 39^{\circ}\text{C}$
- New/increasing tachycardia
- Generalised flushing
- Generalised hives/rash (>two thirds of body)
- Acute onset bleeding diathesis

- DO NOT restart the transfusion; disconnect the giving set.
- Keep line open with new giving set and 0.9% saline.
- Institute resuscitation/supportive care.
- CALL transfusion medicine lab
- Consider MET call.
- Complete transfusion reaction form.
- **SEND TO THE TRANSFUSION LABORATORY**
(Date, time and sign tubes and request form)
 - 1 X pink top tube (EDTA): Grp and screen, crossmatch, DAT
 - 1 X lavender tube (EDTA): FBC, film, reticulocyte count
 - 1 X green top tube (Lithium heparin +gel): UEC, LFT, LDH, haptoglobin
 - **IF suspected sepsis/bacterial contamination** (temp $\geq 39^{\circ}\text{C}$ or temp increase of $\geq 2^{\circ}\text{C}$ above baseline with haemodynamic instability): send blood cultures and implicated product with giving set
 - **IF respiratory distress:** add BNP, CXR.
 - **IF severe allergic/anaphylaxis:** send 1 x gold top (SST) for immunology, tryptase, IgA antibodies.
- Call Haematology for advice, particularly if further transfusion is required.

Serious transfusion reactions include:

- Haemolytic (acute and delayed)
- Serious febrile non-haemolytic
- Severe allergic reaction/anaphylaxis
- Transfusion-related acute lung injury (TRALI)
- Transfusion-associated circulatory overload (TACO)
- Bacterial contamination, ie: septic episode